

# VAHE R. JABAGCHOURIAN

## Hardware Design Engineer

### SUMMARY

Adept technologist and an insatiably inquisitive MSEE degree holder seeking a full-time position as a hardware design engineer.

### ENGINEERING SKILLS

#### Lab Equipment

Digital Oscilloscope, Frequency Spectrum Analyzer, Digital Multimeter, Logic Analyzer

#### Hardware Languages

Assembly, C, C++, Verilog HDL

#### EDA Development / Simulation Tools

Xilinx Vivado, ModelSim, Mentor Graphics ModelSim, Xilinx ISIM, Cadence NCVerilog/NCSim, Xilinx ChipScope

#### Operating Systems

Microsoft Windows, UNIX, Linux, MacOS

#### Productivity Software

Word, Excel, PowerPoint

#### Web Technologies

HTML, PHP, JavaScript, SQL, CSS, ReactJS

### RESEARCH PROJECTS

#### International Digital Terrestrial TV Receiver Design Proposal, and Feasibility Analysis

Independent Study for MS Degree

May 2008 Northridge, CA

Researched and proposed a system-level concept for an international Over-The-Air Digital TV receiver using an FPGA in a feasibility document.

#### Electric Vehicle Battery Equalization

Independent Study for BS Degree

Jan 2003 – Mar 2003 Irvine, CA

UCI Undergraduate Independent Study

Led electric vehicle battery equalization project: Planned, researched equalization, sourced modules, & tested design of a Kewet electric vehicle battery equalization system. Secondly, built a Rube Goldberg contraption, and received 2nd place, and \$150 on 2003 UCI E-Week.

## Electronics Engineer

## Digital Design | Verilog HDL | EDA Tools | Test/Measurement Equipment

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🌐 LinkedIn: <https://www.linkedin.com/in/vahe-jab/> 🏠 Home: West Hills, CA

### EDUCATION



M.S. Electrical Engineering

Aug 2007 – Dec 2011

[California State University, Northridge](#)

Northridge, CA

Master's Project - Implementation of a 3D Graphics Rendering Pipeline Using an FPGA  
<https://scholarworks.calstate.edu/downloads/sb397d249>



B.S. Computer Engineering

Sep 2001 – Mar 2007

[University of California, Irvine](#)

Irvine, CA

Senior Design Project - Voice Activated Presentation Assistant

<https://www.scribd.com/document/129977363/Voice-Activated-Presentation-Assistant>

### UNDERGRADUATE COURSES

Microprocessor Systems  
3D Computer Graphics  
Digital Systems Design in VHDL  
Communication Systems  
Compilers & Interpreters  
Digital Electronics

### GRADUATE COURSES

Verilog HDL for Digital Systems Design  
Computer Architecture  
3D Scientific Visualization  
VLSI Design/Testing  
Digital Systems Verification  
Computer Arithmetic Design

### EXPERIENCE

#### PHP Developer and IT Assistant

Sep 2020 – Jan 2021

[American Business Insurance Services Inc](#)

Westlake Village, CA

- Produced a dynamic sorter in JavaScript for reordering HTML of tabbed report links by name alphabetically reducing report link search time
- Corrected driver ordering scripts in PHP to prevent reorder of driver records 60 days before, resulting in savings of thousands of dollars of extra order cost
- Added ability to highlight excluded drivers across PHP forms, and allowed quick re-query of exclusions on driver changes, using scheduled task
- Created plugged in, unplugged date/time list per vehicle of trackers for vehicles based on vehicle VINs and added time zones to help CSRs (Customer Service Representatives)

#### Web Applications Developer

Jun 2016 – Jan 2018

[Rinsoft LLC](#)

Calabasas, CA

- Created front end-templates for an inventory management system from UI/UX designer using mockups/wireframes within 1 to 2 weeks from inception to completion.
- Implemented C#/ASP.NET/MSSQL Auto Ordering API for automatic order execution, after diagramming PowerPoint flow chart for program logic
- Communicated regular updates of tasks/issues to project manager, developers, stakeholders and finally clients during integration, every day
- Developed REST API integration documentation and high-level computer-aided diagrams for showing API data and logic flow, using draw.io and MS PowerPoint
- Accomplished integration with LightSpeed picking software and VendSys ordering API in 6 months, demoing results to vendor/client

#### Software Developer

Jun 2014 – Jun 2015

[Rinsoft LLC](#)

Calabasas, CA

- Reduced graphic artists' project cycle time via developed CSS/AngularJS/Masonry.js dashboard for optimizing graphic artist project workflow
- Produced a 10 worksheet pivot table for a 10 round 2016 FCC broadcast reverse incentive auction in Excel from a single table  
<https://www.fcc.gov/about-fcc/fcc-initiatives/incentive-auctions/how-it-works>
- Fox Networks Engineering and Operations directly utilized produced deliverables across 3 different departments - scheduling, engineering and graphics

## PERSONAL CHARACTERISTICS

Architectural, systematic, and habitual in planning, performing, and verifying accuracy of work

Quickly able to assimilate new information

Eager and excited to receive corrective feedback for personal modification

Continuously self-aware and self-initiative

Purposeful and driven to accomplish multiple goals

Meticulously, persistently, and relentlessly attentive to small details

Endlessly enthusiastic and energetic

Excellent in computing numbers across technological, schedulable, and total gains/costs

Strongly driven by a sense of time urgency in the matters of immediate priority matters that demand complete attention, resources, accuracy, and timely completion

Charismatic & naturally able to relate to anyone at their own level

Strongly, openly, communicative and articulate in technical, non-technical, and empathetic matters

Naturally inclined and hungry to lead by example, and by service

Persistent, persuasive, and able to bring out the very best in others

Methodically & strategically organized

Insatiably curious and inquisitive about computers, electronics, and technology

Capable of working both independently and within team environments

## LANGUAGES



## HARDWARE PROJECTS

### Voice Activated Presentation Assistant

[University of California, Irvine](#)

Aug 2006 - Mar 2007

Irvine, CA

<https://www.yumpu.com/en/document/view/30414803/voice-activated-presentation-assistant-eeecs-senior-design-project>

- Designed and constructed a Voice Activated PowerPoint Controller incorporating the HM2007 speech recognition board, VersaPoint transmitter, Arduino microcontroller, and countdown display.
- Created a driver program in the C programming language to enable the system's functionality.
- Utilized schematic capture software (EaglePCB) to generate technical computer-aided drawings, illustrating the complete system at both the chip and component level, and these drawings were exported for inclusion in the final project report.
- The final project report served as a high-quality model and was employed by the UCI EECS department to directly support ABET (Accreditation Board for Engineering Technology) accreditation.

### Xilinx Virtex-7 VC707 Based FPGA GPU Implementation

[https://github.com/vahejab/artty\\_sv\\_sampler](https://github.com/vahejab/artty_sv_sampler)

Jan 2023 – Present

West Hills, CA

- Documented Q/A (Jan 2023 – March 2023) available on Electronics Stack Exchange at <https://electronics.stackexchange.com/users/20332/vahe?tab=questions>
- Completed development of PS/2 mouse input module for a 3D Graphics Accelerator in 8 weeks using FPro (FPGA Prototype) template design from Dr. Pong Chu's prototype example in FPGA Prototyping by SystemVerilog examples (Xilinx Microblaze MCS SoC Edition) textbook
- Performed behavioral and timing simulation on ModelSim to identify primary FIFO issues and resolved them for both UART and PS/2 mouse modules
- Conducted synthesis and validation on Xilinx VC707 FPGA Evaluation board while connected to PS/2 mouse and USB UART with host PC
- Used PuTTY terminal to observe real-time mouse movement packets from a 3D trackball across three axes and monitored button presses
- Located an appropriate Level I2C (Inter Integrated Circuit) shifter for bidirectional voltage shifting between +1.8 Volts and +5 Volts
- Assembled the Level Shifter on a Digilent PMOD BB (Breadboard) and connected it to a Digilent, Inc. FMC-CE expansion board, which plugs onto the Virtex-7 evaluation board via one of two FMC Mezzanine connectors

### Analog To Digital Converter

[University of California, Irvine](#)

Aug 2005 – Mar 2006

Irvine, CA

- Designed and simulated a 6-bit Flash Analog to Digital Converter (ADC) using Magic, IRSIM and HSPICE
- Received fabricated ADC chip from MOSIS for testing
- Developed printed circuit board using ExpressPCB layout editor
- Assembled components on circuit board and validated chip functionality using function generator, digital oscilloscope, and a digital logic analyzer

## VOLUNTEER WORK

### Gym Re-Racker

Nov 2021 – Present

Crunch Fitness

Chatsworth, CA

Conducted individual daily, weekly and increasingly frequent and regular equipment re-rack and organization efforts for the direct benefit of the entire facility (management, employees, and members).

Expanded voluntary efforts two other sites (Northridge, CA, and newly acquired West Hills, CA facility - formerly Athletic Society).

Achieved complete re-rack organization and site cleanup for full facility on average of 5 hours on each visit.

Organization efforts resulted in significant and noticeable improvements in facility member reviews/ratings reaching 4/5 stars.

## EXTRA PROJECTS

### Spacecraft Testbed Power Loss Calculator

Individual Project

Jan 2015 - Feb 2015

West Hills, CA

Created a web application capable of allowing engineers to quickly see and calculate power loss in decibels(dB) across any two nodes from cables on electronic assemblies of a spacecraft. Performed application development in HTML, and JavaScript along and produced three local files to providing power levels for parsing.

## WEB APPLICATION PROJECTS

### Risk Management Web Application

2022 – Present

<https://github.com/vahejab/RiskAIM-Overview>

- Produced a single Angular.JS project risk management web application to track and reduce project risks via REST API-centric architecture
- Updated controllers to a component-based framework in ReactJS and upgraded RESTFUL API to a feature-based vertical slice architecture / application directory structure
- Achieved creation of an advanced adaptive tracking form to track risk mitigation events using waterfall metrics, and PowerPoint reports with a 5 by 5 risk matrix showing risk level mitigation progress
- Completed development of a data visualization dashboard grid using Gridster.js (Front End JavaScript Grid Library), Angular-Gridster.js, d3.js (Data Driven Documents), and dc.js (Dimensional Charting) library
- Utilized HTML5 for dropdowns, date picker for calendar control, and pre-existing sample tabular code for producing responsive tables
- Accomplished construction of visualization mockup for dashboard data visualization of entire project risks (specifically buildup, burndown, and bar-chart metrics)
- Incorporated PHP (Hypertext Preprocessor) Composer library for dependency management as well as latest features of PHP8 for annotation-based REST (Representational State Transfer) URL (Uniform Resource Locator) routing
- Leveraged PHP OpenTBS (Tiny But Strong) for report generation, and PHPOffice for generation of PowerPoint graphs, 5 by 5 risk matrix, and a risk mitigation waterfall chart

### Action Item Management Web Application

2013 – 2017

[https://www.ieee-bv.org/wp-content/uploads/2016/07/newsletter\\_2016\\_07.html#b4](https://www.ieee-bv.org/wp-content/uploads/2016/07/newsletter_2016_07.html#b4)

- Created an action item management app in PHP/MySQL along with D3.js dashboard status charts (build-up, burn-down, stacked bar)
- Incorporated sortable summary page with red colored indicators on dates for past due action items and allowed item searching/filtering
- Programmed functionality to activate, schedule, and send automatic email reminders of past due and approaching action items
- Delivered a presentation/demonstration for IEEE (Institute of Electrical and Electronics Engineers) Buenaventura Section
- Demonstrated complete functionality of application which was loaded onto an AWS (Amazon Web Service) hosting environment configured for public viewing